

# Samyukta Suryanarayanan

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## Education

### Columbia University, Columbia Engineering

*M.S. in Computer Science* | Track: *Machine Learning*

Classes: AI for software Security, Machine Learning, Data Science, Deep Learning, Probability & Statistics

New York, NY

Available Full Time

### Rutgers University Honors College

*B.S. in Computer Science* | Minors: Business Administration, Mathematics

Camden, NJ

## Technical Skills

**Full-Stack Development:** React 18, Django 5.0, Flask, RESTful APIs, Tailwind CSS, Vite

**Data & Databases:** PostgreSQL, MySQL, DynamoDB, Data Normalization, Schema Design, Data Modeling

**Cloud & Infrastructure:** AWS (S3, SQS, EC2, Bedrock), Docker, Docker Compose, Redis, GitHub Actions (CI/CD)

**AI/ML Engineering:** Python, PyTorch, Scikit-learn, LangChain, RAG, Multi-Agent Orchestration

**Tools:** Git/GitHub, VS Code, Agile, Technical Risk Reporting, Unit Testing

**AI Dev Tools:** Claude, Copilot, Gemini, Grok

## Projects

### *Global Regulatory Compliance Dashboard*

- Architected a production-grade full-stack platform using React 18 and Django 5.0, orchestrating a 5-service containerized environment with Docker Compose to streamline the analysis of global regulatory data.
- Engineered an asynchronous data pipeline using Celery and AWS SQS to ingest and normalize 10k+ semi-structured OpenFDA records, implementing an immutable AWS S3 data lake and field-level audit logs to ensure 100% data traceability for quality auditors.
- Developed a Scikit-learn risk-scoring heuristic to quantify facility hazards (0.0–1.0), integrating predictive insights into a custom React dashboard featuring Recharts for trend visualization and advanced server-side filtering.

### *Real-time Transaction Anomaly Detection System*

- Engineered an event-driven risk monitoring system using a Flask API and Redis queue to ingest and score financial transactions with low-latency.
- Established a model governance framework by integrating a PyTorch autoencoder that persisted reconstruction errors, thresholds, and versioning to ensure 100% auditability and reproducibility of risk-based decisions.
- Strengthened data integrity controls by implementing tenant-safe data isolation and idempotent writes, effectively mitigating risks of data leakage in enterprise environments.

### *SSRF Auditor*

- Developed a hybrid security pipeline for detecting Server-Side Request Forgery (SSRF) in AI-generated code, integrating static analysis with multi-agent LLM triage to improve detection precision.
- Leveraged Claude and multi-agent LLM orchestration to automate SSRF vulnerability triage, reducing manual review time and achieving a projected 40% reduction in false positives through trace-based grounding.
- Architected an iterative automated patching module that generates security remediations in a sandboxed environment, achieving a predicted 50%+ success rate through closed-loop feedback.

## Experience

### *Computer Science Tutor*

May 2024–Present

- Mentored students in designing multifile Python, Java, and SQL projects, emphasizing modular architecture, reproducible setups, and maintainable code structure.
- Taught debugging, unit testing, and deterministic testing practices to reinforce correctness and reliability in real-world engineering workflows.
- Guided learners in documenting system behavior, writing setup instructions, and structuring development environments, mirroring professional engineering standards.